

CISC 667

Introduction to Autonomous Driving

Lecture 6
Oct 13, 2025

Today's Outline

- Localization
 - Fundamentals
 - Existing Approaches
- Mapping
 - Fundamentals
 - Existing Approaches
- SLAM

Demo Video

<https://user-images.githubusercontent.com/21349875/219626075-d67e9165-31a2-4a1b-8c26-9f04e7d195ec.mp4> (KISS-ICP)

<https://www.youtube.com/watch?v=luBGKxgaxS0> (ORB-SLAM)

Localization and Mapping

- What technologies do we have?
- High-frequency pose tracking
- Low-frequency global re-localization

Fundamentals

- Human's logical process. Where am I?

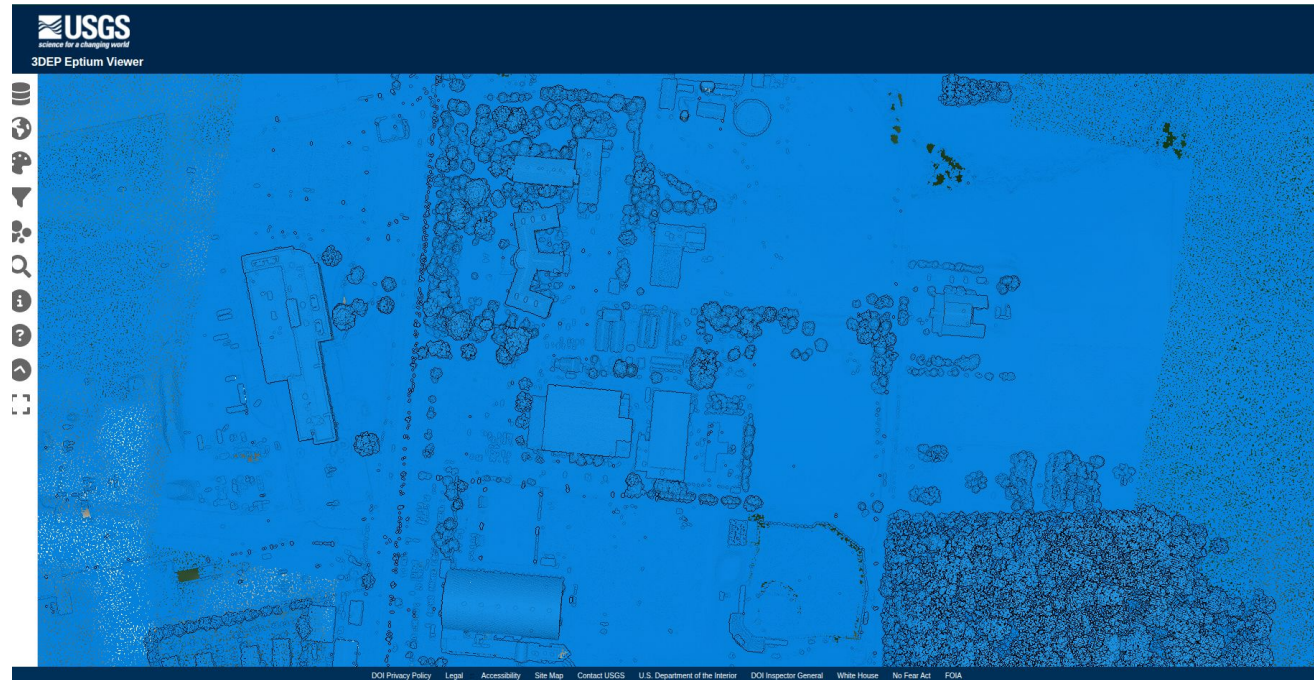


Translate human thinking into code

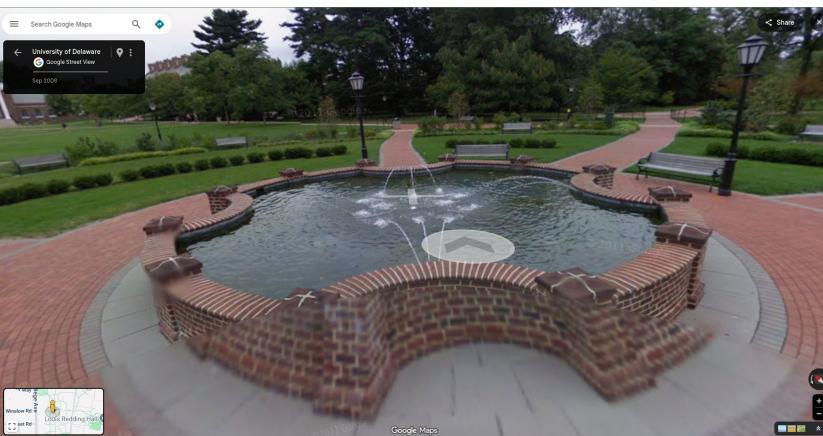
- I have some sort of pre-built map (images in this case)
- Get a rough sense of where I might be
- Search through all available possibilities
- Output final location and orientation guess

Other sensors?

- Where am I?



Mapping



What is the change in my location/orientation?

Localization

- ICP - Iterative Closest Point
- NDT - Normal Distribution Transform

Limitation

- Camera operates at 30Hz, LiDAR operates at 10Hz. This may be a problem when...
- A spontaneous calculation error give me an estimated location on the other side of the world, what can i do?
- Other problems?

Example

https://github.com/NaoakiAkai/icp_ndt_python

Lanelet2 Map